The f-divergence is a family of divergences between two probability distributions p and q defined by a convex function $f : \mathbb{R}_+ \to \mathbb{R}$. It is defined as

$$\mathbb{D}_f(p\|q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right)\, dx$$

The function f is called the generator of $\mathbb{D}_f$, and is always convex. The f-divergence is a generalization of the more-famous Kullback-Leibler divergence, which corresponds to the case where $f(t) = t \log t$.

0.0.1 See also

0.0.2 References